



GANESH REDDY

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Portfolio | LinkedIn | GitHub

Professional Summary

M.Sc. Autonomy Technologies student at FAU Erlangen–Nürnberg with hands-on experience in **sensor-based robot guidance, industrial image processing, and robot interface development**. Proficient in **Python, C++, ROS2, MoveIt, and OpenCV**, with practical project work spanning perception-to-action pick-and-place systems, 5-DOF arm kinematics, and SLAM-based navigation. Experienced in designing and conducting structured experiments on robot systems and deriving optimized, reproducible solutions, eligible for FAU mandatory internship (20+ weeks, available from immediately) and also for working student role.

Education

Friedrich-Alexander University (FAU)

M.Sc. Autonomy Technologies

Erlangen, Germany

Apr 2024 – Present

- **Relevant Coursework:** Robotics in Surgery & Diagnostics, Human-Robot Co-adaptation, AI in Medical Robotics, Control Systems, Robot Kinematics, C++ Fundamentals, Robotics 1, Robotics 2
- **GPA:** 2.7 | **ECTS:** 67.5 / 120
- Programme includes a mandatory 10-week industrial internship (FAU, Feb 2024)

KLE Technological University

B.E. Automation and Robotics

Hubballi, India

Aug 2018 – Jul 2022

- **Focus Areas:** Robot Kinematics, Control Systems, Machine Design, Sensor Fusion
- **GPA:** 2.3 | **Thesis:** Automated Grading & Sorting System for Vegetables and Fruits (Python, OpenCV, SolidWorks)

Professional Experience

Friedrich-Alexander-Universität Erlangen–Nürnberg (FAU)

Student Research Assistant — Part-time

Erlangen, Germany

Nov 2025 – Present

- Develop **Python-based automated evaluation scripts** for research datasets, applying structured data-validation in an academic R&D environment; improved turnaround by ~20%.
- Clarify Python and data-workflow questions for students, reducing clarification cycles by ~25% via systematic debugging examples and documentation.

Digital Associates India

Internet of Things (IoT) Intern

Hyderabad, India

Jan 2023 – Oct 2023

- Developed **Python/C++** sensor data acquisition pipelines for ESP32-based systems; applied BLE/RSSI environmental sensing principles for indoor localisation.
- Designed **PCB schematics/layouts (KiCad/Eagle)**; performed hardware testing and debugging with oscilloscope and multimeter, reducing rework by ~20%.
- Documented test setups and Agile milestones in Jira; supported iterative firmware–hardware integration.

Centre for Autonomous Systems Engineering

Systems Engineer Intern

Hubballi, India

Jan 2022 – Jul 2022

- Designed and modelled a **5-DOF robotic arm** in SolidWorks with Dynamixel XM430 servos; analysed joint kinematics, mechanical tolerances, and motion constraints.
- Built **URDF/Xacro models** and validated kinematics and motion sequences in **Gazebo**; integrated **MoveIt** for trajectory planning using simulation-first methodology.
- Performed iterative motion parameter tuning; documented failure modes and test procedures for team reuse; wrote Python scripts to parse simulation logs and evaluate sequences.

Technical Competencies

- **Robotics & Autonomy:** ROS/ROS2, Gazebo, RViz, MoveIt, URDF/Xacro, Nav2, AMCL, SLAM, TurtleBot
- **Programming:** Python, C++, C, MATLAB
- **Sensor & Image Processing:** OpenCV, YOLOv3, CNNs, BLE/RSSI, IMU, LiDAR
- **Simulation & Modelling:** Gazebo, MuJoCo, MATLAB/Simulink, URDF modelling

- **Control Engineering:** PID control, motion planning, trajectory execution, kinematics
- **ML & Vision:** PyTorch, TensorFlow, OpenCV, YOLOv3, CNNs
- **CAD/Mechanical:** SolidWorks, Autodesk Inventor, 3D modelling, technical drawings
- **Tools:** Git, Linux (Ubuntu), VS Code, Jira, Power BI, Microsoft Office

Projects

Franka Panda Pick-and-Place — Robot Manipulation & Perception-to-Action

- Developed a perception-to-action pipeline integrating color-based object detection with structured robot motion execution (approach, grasp, place), directly analogous to sensor-based robot guidance for joining, searching, and grasping tasks.
- Validated task repeatability through iterative tuning of motion and task-sequencing parameters.
- *Tools:* Python, ROS2, MoveIt2, OpenCV.

5-DOF Robotic Arm — Simulation, Kinematics & Motion Planning

- Designed arm in SolidWorks with Dynamixel XM430 servos; built URDF/Xacro model and validated joint kinematics and trajectories in Gazebo & MuJoCo.
- Integrated MoveIt for motion planning; performed iterative parameter tuning and documented failure modes for team reuse.
- *Tools:* SolidWorks, ROS2, Gazebo, MuJoCo, MoveIt2, Python.

Real-Time Face and Object Detection (Computer Vision)

- Built a Python-based multi-threaded detection pipeline using **YOLOv3** and Haar Cascades; optimised detection latency through multithreading and parameter tuning.
- Validated robustness under varied lighting conditions using OpenCV on Linux.
- *Tools:* Python, PyTorch, OpenCV, YOLOv3, Haar Cascades, Linux.

SLAM & Navigation with TurtleBot (Mobile Robotics)

- Developed a simulation-based validation pipeline for autonomous navigation: extracted scenario configurations (multi-environment YAML), ran systematic simulation tests in Gazebo, and evaluated map consistency and path performance across multiple runs, directly analogous to scenario-based driver assistance validation.
- Configured and tuned Nav2 parameters to define and test new scenario types, mirroring the process of generating simulation-based scenarios for function validation.
- *Tools:* ROS2, Nav2, SLAM, AMCL, Gazebo, LiDAR, IMU, RViz.

Automated Grading & Sorting System — Computer Vision & Robotics

- Led a **5-member team** to design a conveyor-based sorting system in SolidWorks; built a Python/OpenCV classification pipeline achieving **80% accuracy**.
- Coordinated sensor-actuator integration in ROS; documented failure modes to improve repeatability and robustness.
- *Tools:* SolidWorks, Python, OpenCV, ML, MATLAB, ROS, Linux.

Certifications

- | | |
|---|-----------------|
| • ROS for Beginners I & II — SLAM, Navigation, Path Planning | <i>Udemy</i> |
| • Python for Everybody — University of Michigan | <i>Coursera</i> |
| • Capstone: Retrieving, Processing & Visualising Data with Python | <i>Coursera</i> |
| • Full Stack Web Development | <i>Udemy</i> |

Languages

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|----------------|-------------------------|
| English | C1 – Fluent |
| German | A2 – Actively improving |
| Hindi | Native (C1) |

Availability: Immediately / FAU mandatory internship eligible (10+ weeks) / Work model: On-site / Hybrid / Hours: 15–20h/week (working student) or full-time (internship / thesis)